

Improving Voltage Ride-Through Procedures in Distributed Generation Systems by Reinforcement Learning

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Abstract. During the last two decades, the operation of the electrical grid has undergone significant changes. This evolution is closely tied to the integration of power electronics into distributed generation systems, which led to increased utilization of renewable energy and as a consequence mitigating climate change by lowering emissions. On the other hand, artificial intelligence plays a considerable role in shaping the development and progress of various technologies, such as the electrical grid. This work presents a study for the application of Reinforcement Learning (RL) tools in distributed generation systems. The objective of using RL is to address voltage perturbations in real time. RL will be useful to maximize the resilience of the system in the event of short circuits of a short duration and minimize the risk of disconnect.

Keywords. AI Applications, Agents and Multi-agent Systems, Hybrid systems

1. Introduction

The reduction in carbon emissions strongly influences the operation of the electrical grid [1]. Generation moves from large production plants to small and medium distributed generation sources. Power electronics is a key factor to achieve this transition. These electronic devices, present in photovoltaic panels and wind turbines, are inherently different than large synchronous generators that have been powering the electrical network in the past. In particular, they have very fast response time and have almost no inertia.

Short circuits are one of the most important electrical systems disturbances because they can cause a general blackout [2]. Grid operators have forced distributed generation systems to help alleviate the effects of these disturbances by using grid codes [3]: the production plant must remain connected while waiting for the protection relays to isolate

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the area causing the problem, if the protection mechanisms are not effective, or the disturbance is located at a very close point, then it is possible to disconnect as a protection mechanism, otherwise it must remain connected to the network, this is what is known as low/high voltage ride-through. Network operators must help recover the system by providing voltage support, which involves injecting reactive current proportional to the measured voltage. To improve voltage support strategies, [4] proposed using the grid code to respond promptly to disturbances such as short circuits. Even though this approach offers remarkable advantages, the selection of the strategy solely and exclusively follows compliance with a regulation defined in the grid code, which is static, off-line and does not adapt to changing situations. The state of the art in this area is divided into optimization-based approaches and data-driven approaches. Optimization approaches are accurate, but their dependence on precise system models limits their effectiveness in complex networks. In addition, solving this problem at each time step can be time-consuming in large network sizes. On the other hand, with the progress of simulation tools and improving the computational capacities of GPUs, the data-driven or AI-based model attracted more attention in the research and industrial communities [5].

2. Reinforcement Learning Approach

Reinforcement Learning provides a powerful framework for learning sequential decision-making tasks in a dynamic environment with a lot of uncertainties. In RL, an agent is responsible for making decisions and taking specific actions at each time step a_t . In response to the action, the environment state changes from s_t to s_{t+1} and provides a scalar signal known as the reward r_t [6]. The interaction between agent and environment will continue until reaching to the stopping criteria. The main goal of the agent is to find an optimal policy $\pi(a|s)$, or strategy to achieve the highest amount of reward.

In this study to manage voltage fluctuations in a power system, reactive current should be injected in a phase experiencing high or low voltage. The key elements to define in order to model this problem within the RL framework are:

- **Environment:** It is simulated in a Simulink model and then exported as an FMU (Functional Mock-up Unit). The FMU output is integrated with OpenAI Gym.
- **Observation:** Previous action and current voltage have been considered as observations to represent the state of the environment, $s_t = (v_t, a_{t-1})$. Therefore, the observation space is considered to be a continuous 2×2 box.
- **Action:** The agent can compensate voltage changes by injecting reactive current. The action space considered is limited in the range $-I_{max} \leq a_t < I_{max}$. If a specific phase is experiencing over-voltage, the reactive current or action that should be applied is within the range $-I_{max} \leq a_t < 0$. Conversely, if the phase is tolerating low-voltage, the action should be bounded in the range $0 \leq a_t \leq I_{max}$.
- **Reward:** The reward considered as a voltage correction made by the agent. Three scenarios have been defined: low voltage, $v_{grid} < 0.85$, high voltage, $v_{grid} > 1.1$ and normal case, $0.85 \leq v_{grid} \leq 1.1$.

$$Reward = \begin{cases} v_{inverter} - v_{grid} & v_{grid} < 0.85 \\ v_{grid} - v_{inverter} & v_{grid} > 1.1 \\ -(|v_{grid} - v_{inverter}|) & 0.85 \leq v_{grid} \leq 1.1 \end{cases}$$

3. Reinforcement Learning Algorithm

Several RL algorithms have been employed in the past to balance voltage regulation and reactive power utilization: DQN, PPO [7], DDPG [5]. These are model-free models, which means that they do not require us to predict the dynamic of an environment, reward, and transition function. In this study, Proximal Policy Optimization (PPO) algorithm has been used because for stability and efficiency.

The simulation time in training is considered to be 0.7 seconds. During this period, the system encounters two faults, an under-voltage fault at 0.21 s and an over-voltage fault at 0.42 s. The PPO algorithm from the Stable Baselines3 library in Python has been trained for 100,000 time steps. The time step considered is based on voltage measurement intervals, which occur every 0.01 seconds. In Figure 1, the cumulative reward for one episode is shown with respect to different policies, each trained based on varying numbers of time steps. Based on the training process, the policies trained for less than 6500 time steps focused primarily on exploring the environment through random actions. Afterward, agent starts the learning process and the parameters are optimized to maximize the accumulative reward. The cumulative reward finally converges at around 90,000 time steps. It is worth mentioning that since the problem is continuous and does not have a clear endpoint, the training process finishes after reaching the specified number of time steps.

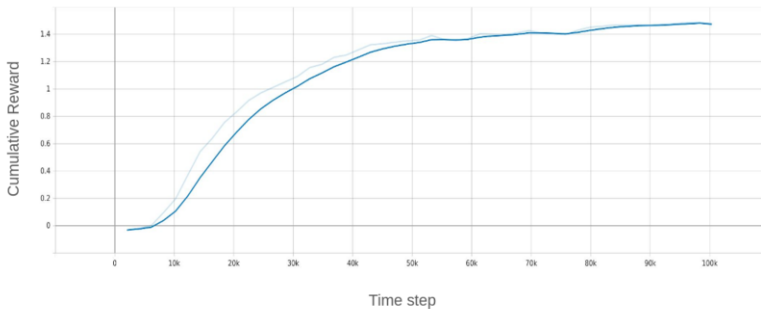


Figure 1. Evolution of reward for 100k time steps

To evaluate the performance of the trained policy over 100,000 time steps, its effectiveness in selecting actions based on different observations and receiving rewards has been analyzed in Figure 2 and 3. In conditions where the system tolerates low voltage, the policy injects the highest amount of leading reactive current into the system and can enhance the inverter voltage compared to the grid voltage. In contrast, under high-voltage tolerance, the policy injects the highest amount of lagging reactive current and decreases the inverter voltage. In both cases, the agent achieves the maximum possible reward of 0.06. Under normal voltage conditions, the agent provides zero reactive current during the first 0.2 seconds and the last 0.1 second, resulting in zero reward, which is optimal in normal case. Moreover, when the system transitions between different voltage scenarios, the agent receives the lowest reward. This variability in reward highlights the challenges that the agent faced while adapting to changes in system conditions.

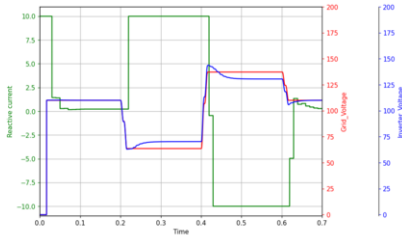


Figure 2. Voltage fluctuations after injecting reactive current

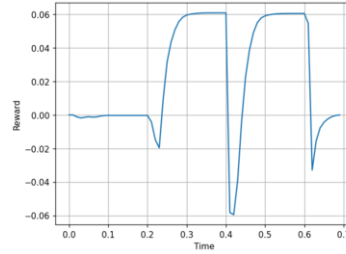


Figure 3. Reward based on the action in each 0.01 second in simulation time

4. Conclusion

This paper presents preliminary work for introducing Reinforcement Learning tools into the voltage ride-through operation of distributed generation systems. Actually, the procedures to support the grid during disturbances are static and off-line, therefore they are not optimized to variable conditions. The proposed PPO based control strategy could capture the time that fault happened in the real-time and produce an appropriate action to deal with the fluctuations in the voltage. Further research is needed to improve the model and achieve higher reward.

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